

Untitled

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```
print(uuid::UUIDgenerate())
```

```
## [1] "e80951fe-a067-4544-941a-493e2f0b3c66"
```

Data

```
commute <- c(18, 25, 32, 41, 27, 35, 22, 19, 45, 38, 29, 24,  
            31, 36, 40, 26, 33, 28, 21, 44, 30, 37, 23, 39)
```

q1

```
minimum <- min(commute)  
maximum <- max(commute)  
range_value <- maximum - minimum  
class_width <- ceiling(range_value / 5)
```

```
minimum
```

```
## [1] 18
```

```
maximum
```

```
## [1] 45
```

```
range_value
```

```
## [1] 27
```

```
class_width
```

```
## [1] 6
```

```

breaks <- c(17.5, 23.5, 29.5, 35.5, 41.5, 47.5)

class_labels <- c("18-23", "24-29", "30-35", "36-41", "42-47")

commute_class <- cut(commute, breaks = breaks, labels = class_labels,
                     include.lowest = TRUE)

frequency <- as.vector(table(commute_class))
cumulative_frequency <- cumsum(frequency)

frequency_table <- data.frame(
  Class = class_labels,
  Frequency = frequency,
  Cumulative_Frequency = cumulative_frequency
)

frequency_table

```

```

##   Class Frequency Cumulative_Frequency
## 1 18-23         5             5
## 2 24-29         6            11
## 3 30-35         5            16
## 4 36-41         6            22
## 5 42-47         2            24

```

The minimum is 18 minutes, the maximum is 45 minutes, and the range is 27 minutes. The class width is 6 minutes.

q2

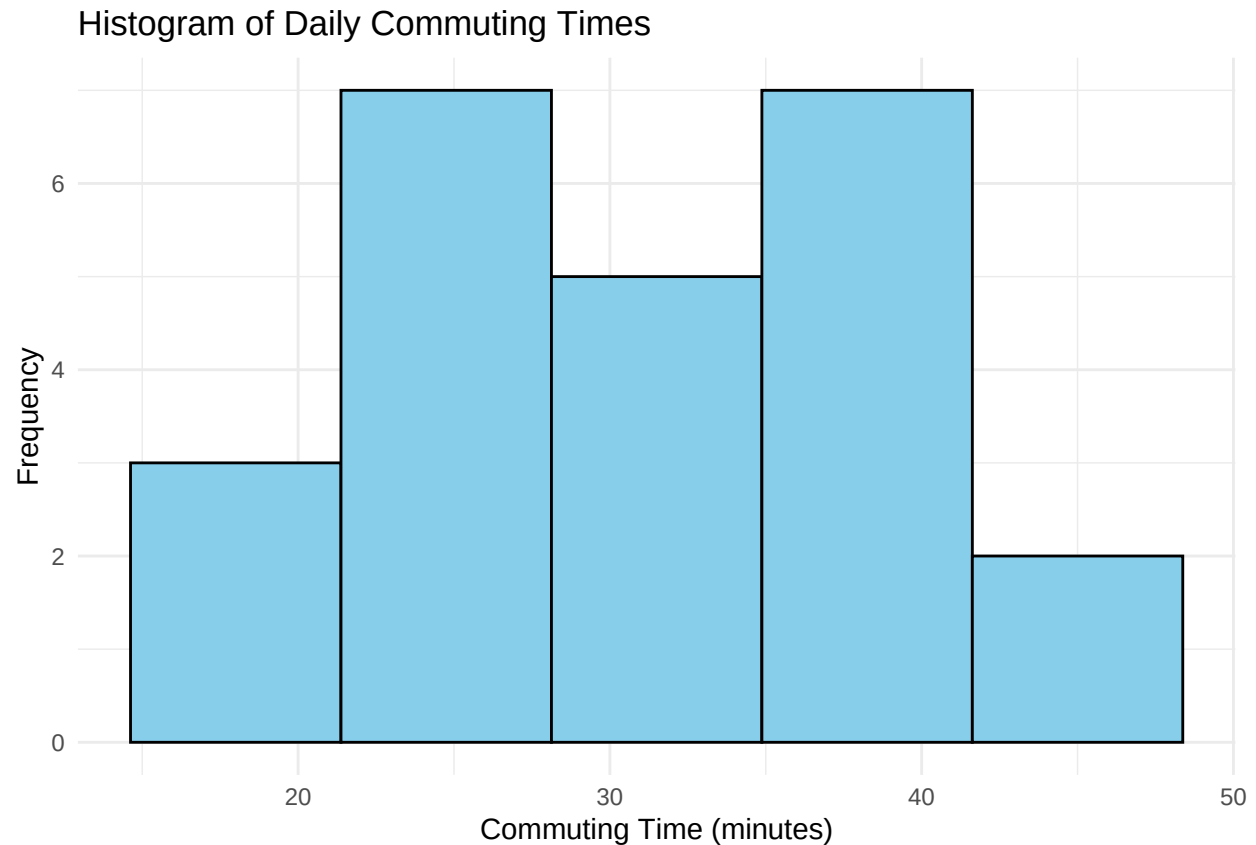
```

library(ggplot2)

histogram_data <- data.frame(commute = commute)

ggplot(histogram_data, aes(x = commute)) +
  geom_histogram(
    bins = 5,
    fill = "skyblue",
    color = "black"
  ) +
  labs(
    title = "Histogram of Daily Commuting Times",
    x = "Commuting Time (minutes)",
    y = "Frequency"
  ) +
  theme_minimal()

```



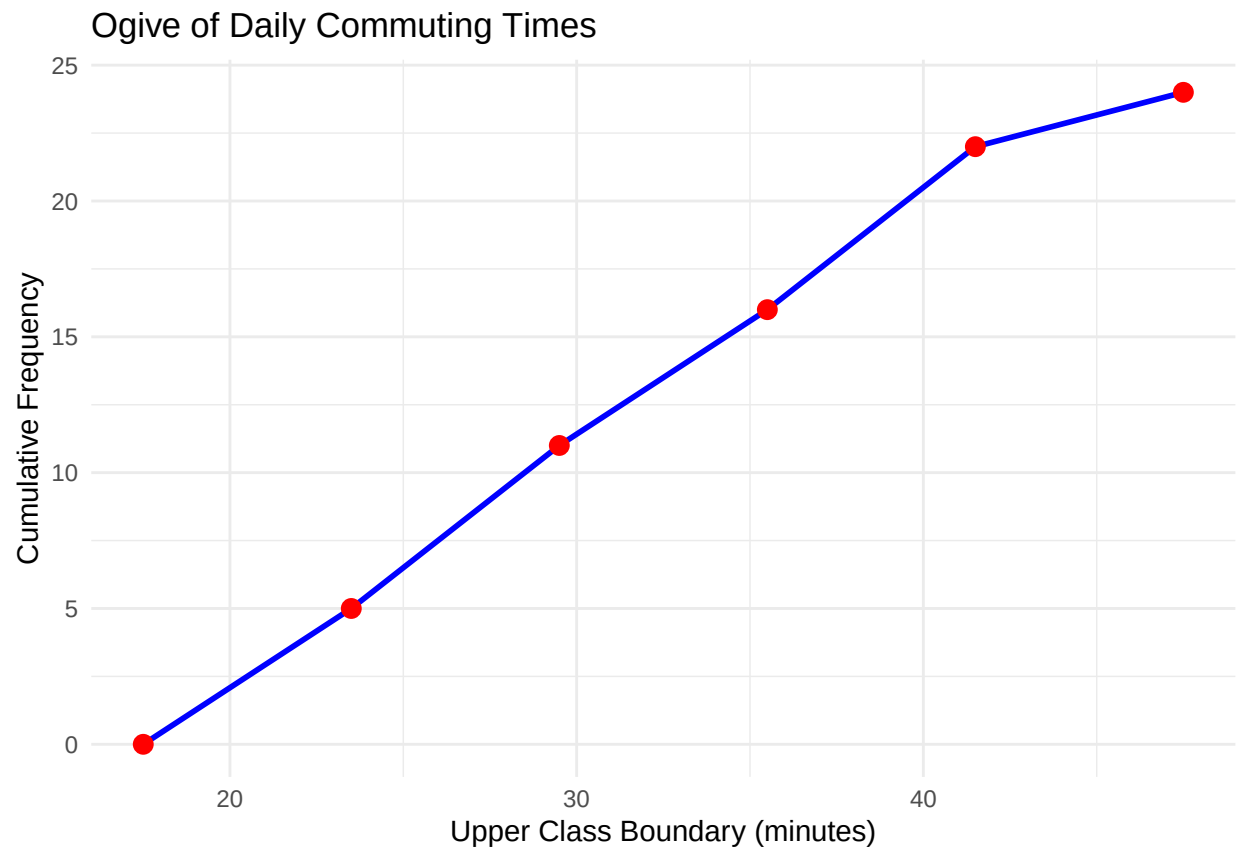
The distribution is approximately uniform, with commuting times spread fairly evenly across the range.

q3

```
upper_boundaries <- c(23.5, 29.5, 35.5, 41.5, 47.5)

ogive_data <- data.frame(
  Class_Boundary = c(17.5, upper_boundaries),
  Cumulative_Frequency = c(0, cumulative_frequency)
)

ggplot(
  ogive_data,
  aes(x = Class_Boundary, y = Cumulative_Frequency)
) +
  geom_line(color = "blue", linewidth = 1) +
  geom_point(color = "red", size = 3) +
  labs(
    title = "Ogive of Daily Commuting Times",
    x = "Upper Class Boundary (minutes)",
    y = "Cumulative Frequency"
  ) +
  theme_minimal()
```



```
estimated_students <- round(  
  approx(  
    x = ogive_data$Class_Boundary,  
    y = ogive_data$Cumulative_Frequency,  
    xout = 35  
  )$y  
)  
  
estimated_students
```

```
## [1] 16
```

From the ogive, approximately 16 students commute less than 35 minutes.